ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems.

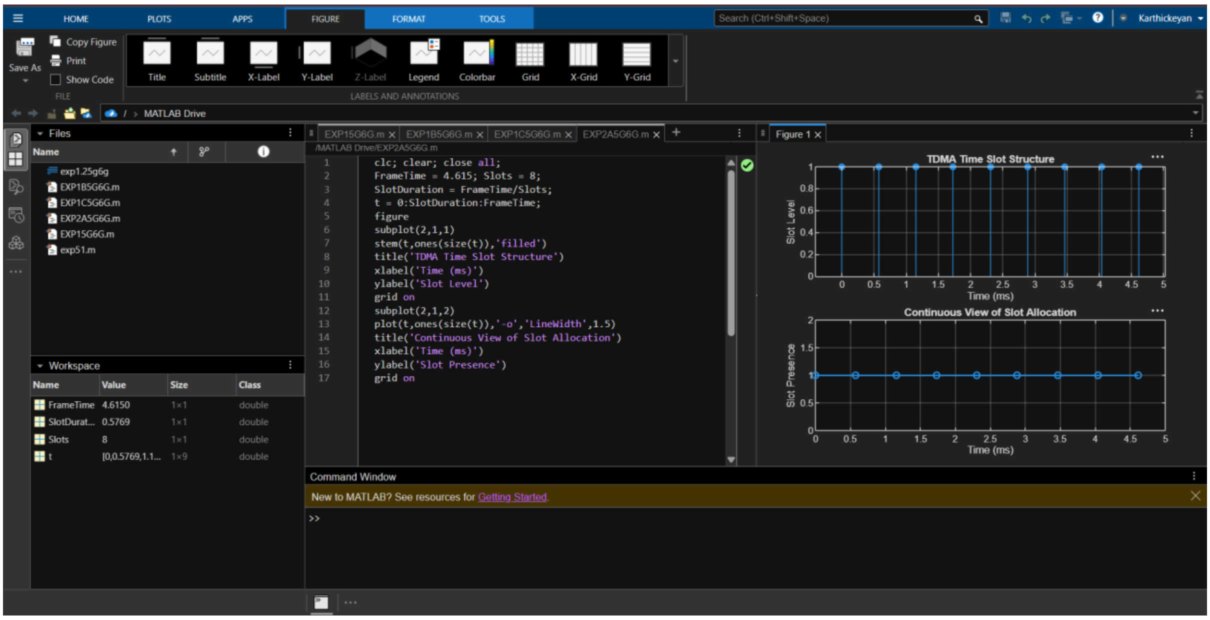
Ex.No. 2: Implementing TDMA Slot Allocation in GSM. Date

Objective 1: Matlab Code and Output:

Program:

```
clc; clear; close all;
FrameTime = 4.615; Slots = 8;
SlotDuration = FrameTime/Slots;
t = 0:SlotDuration:FrameTime;
figure
subplot(2,1,1)
stem(t,ones(size(t)),'filled')
title('TDMA Time Slot Structure')
xlabel('Time (ms)')
ylabel('Slot Level')
grid on
subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5)
title('Continuous View of Slot Allocation')
xlabel('Time (ms)')
ylabel('Slot Presence')
grid on
```

OUTPUT:



Objective 2: Matlab Code and Output

Program:

```
clc; clear; close all;
TotalSlots = 8;
AllocatedSlots = 6;
FreeSlots = TotalSlots - AllocatedSlots;
Utilization = (AllocatedSlots/TotalSlots)*100;
disp('Frame Utilization (%)')
disp(Utilization)
figure
% Graph 1: Simple Line Plot (SAFE - no bar function)
subplot(2,1,1)
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
title('GSM Frame Utilization (Allocated vs Free)')
xlabel('Slot Type Index (1=Allocated, 2=Free)')
ylabel('Number of Slots')
```

grid on

% Graph 2: Pie Chart (SAFE)

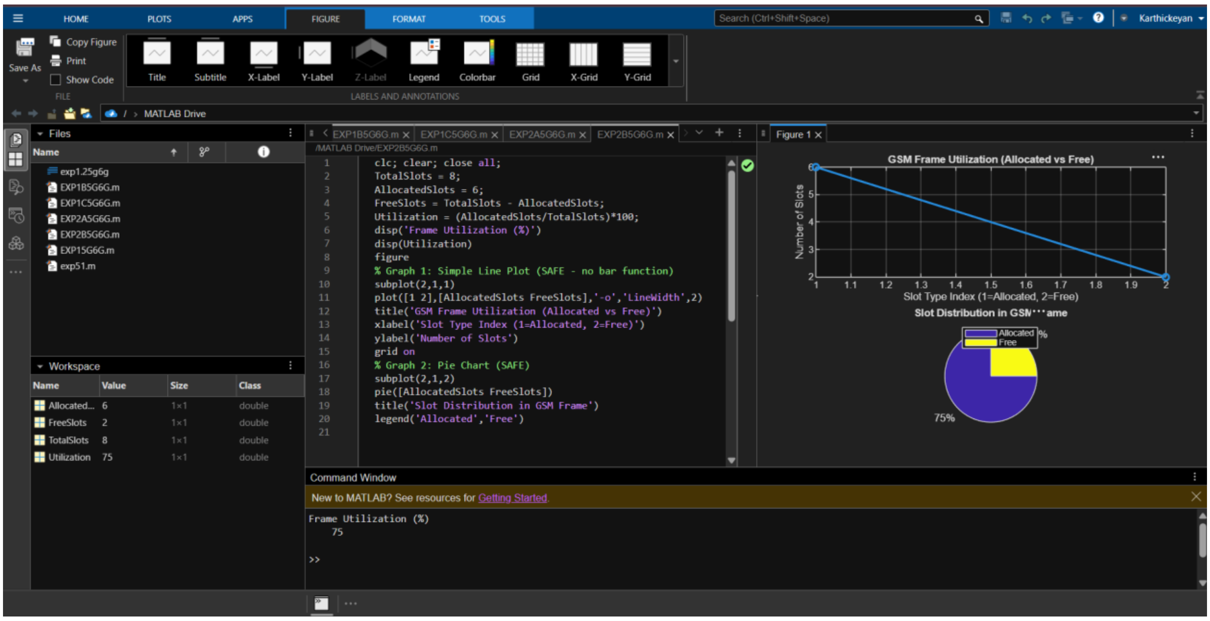
subplot(2,1,2)

pie([AllocatedSlots FreeSlots])

title('Slot Distribution in GSM Frame')

legend('Allocated','Free')

OUTPUT:



Objective 3: Matlab Code and Output

Program:

clc; clear; close all;

u = 1:8;

a = [ones(1,5) zeros(1,3)];

b = 1 - a;

disp(table(u',a','VariableNames',{'User','Allocated'}))

subplot(2,1,1)

stem(u,a,'filled')

```
title('TDMA Slot Allocation')
xlabel('Users')
ylabel('Status')
grid on
subplot(2,1,2)
plot(u,a,'-o',u,b,'-s')
legend('Allocated','Blocked')
title('TDMA Slot Utilization')
xlabel('Users')
ylabel('Status')
grid on
```

OUTPUT: